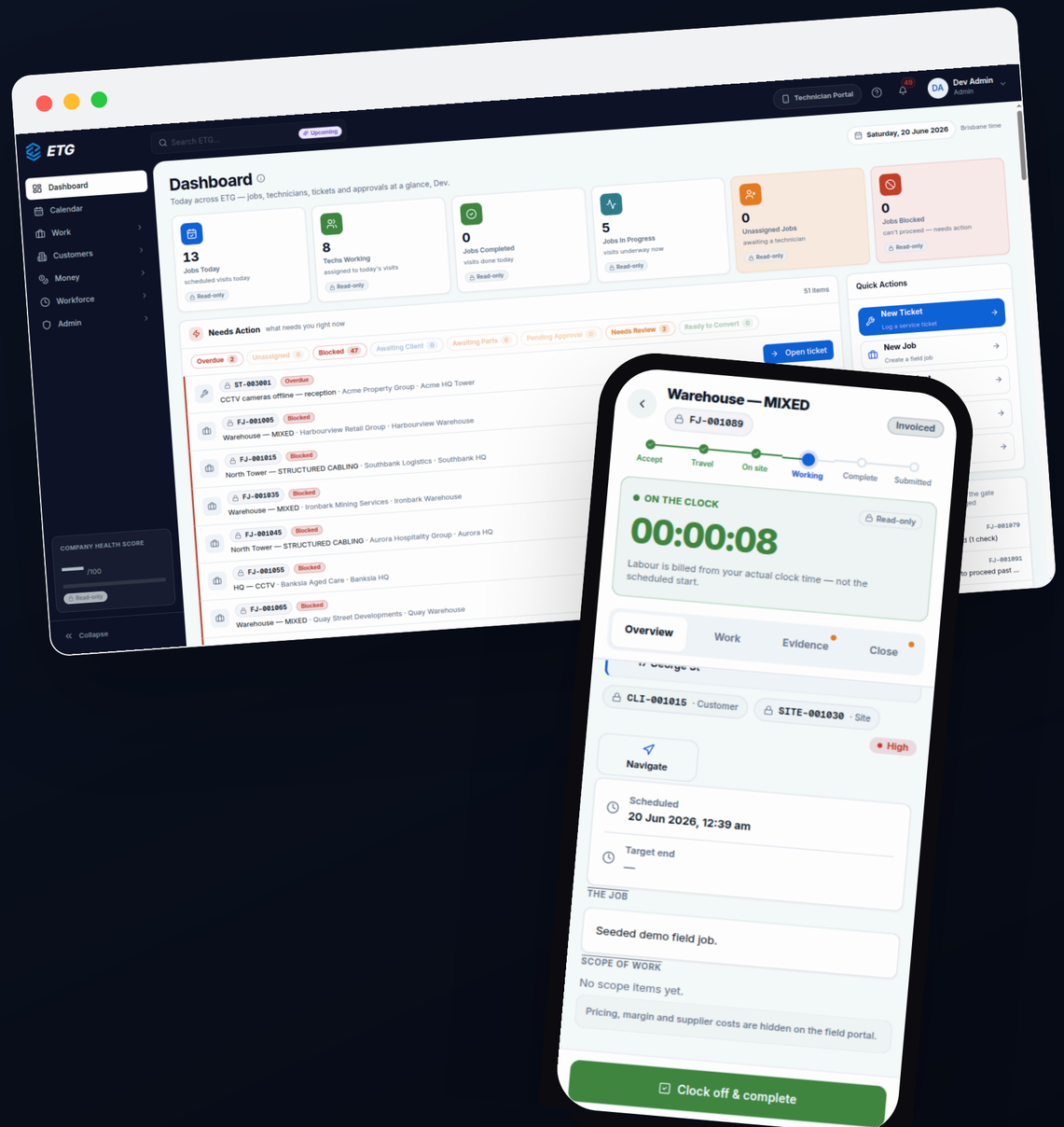
EVOLUTION
DATA & COMMUNICATIONS

Enforcement-First Field-Service OS

An enforcement-first operations platform for an electrical, security, AV & telecom group — dispatch, workforce, assets and dual-sided billing behind one decision engine.



Zohaib Ali Mughal · Sole architect, full-stack

EVOLUTION
DATA & COMMUNICATIONS

01 — OVERVIEW

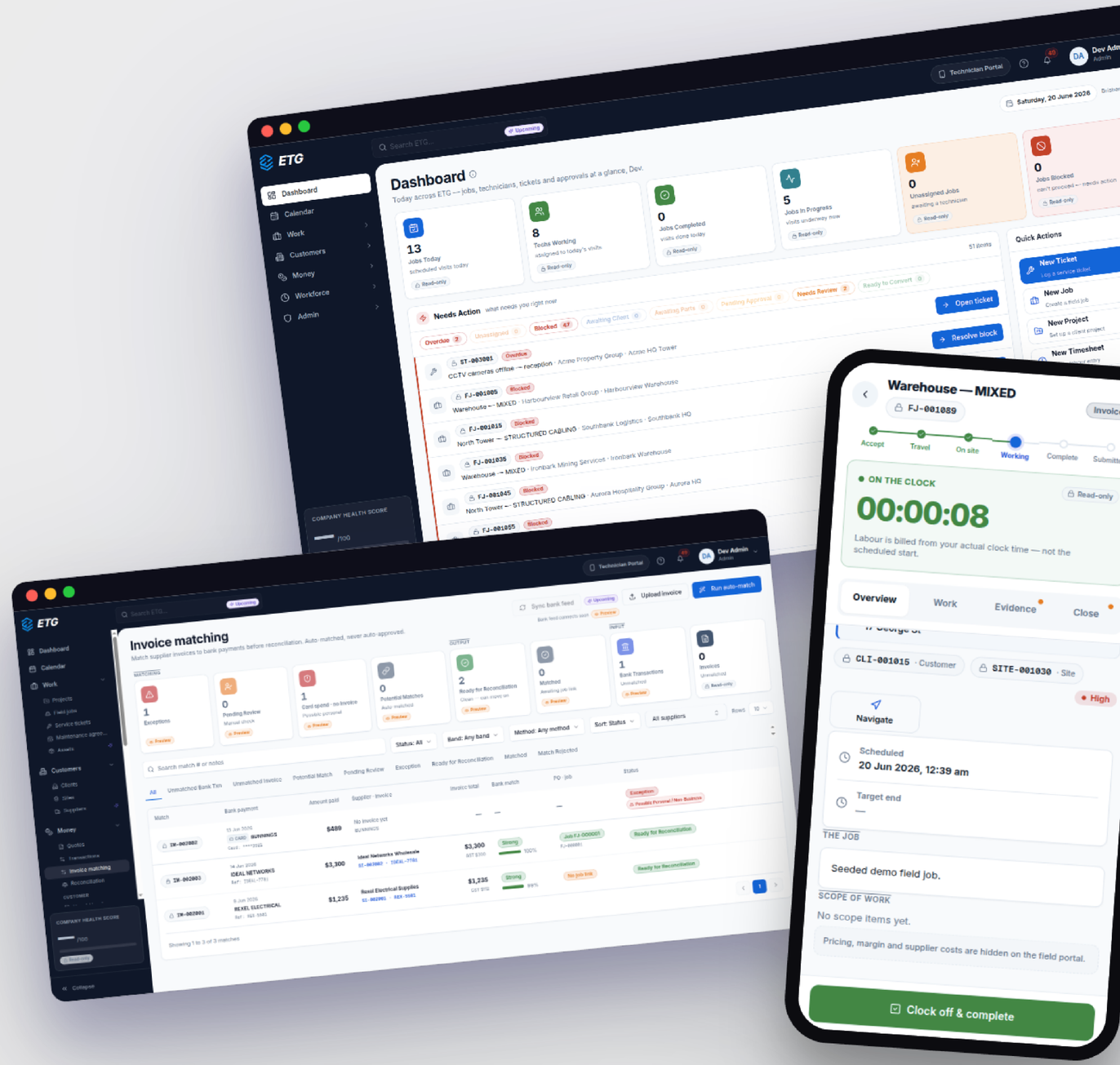
Enforcement-First Field-Service OS

An enforcement-first operations platform for an electrical, security, AV & telecom group — five portals over one decision gate, spanning quote-to-cash, procurement, workforce and billing.

ROLE **Sole architect, full-stack**

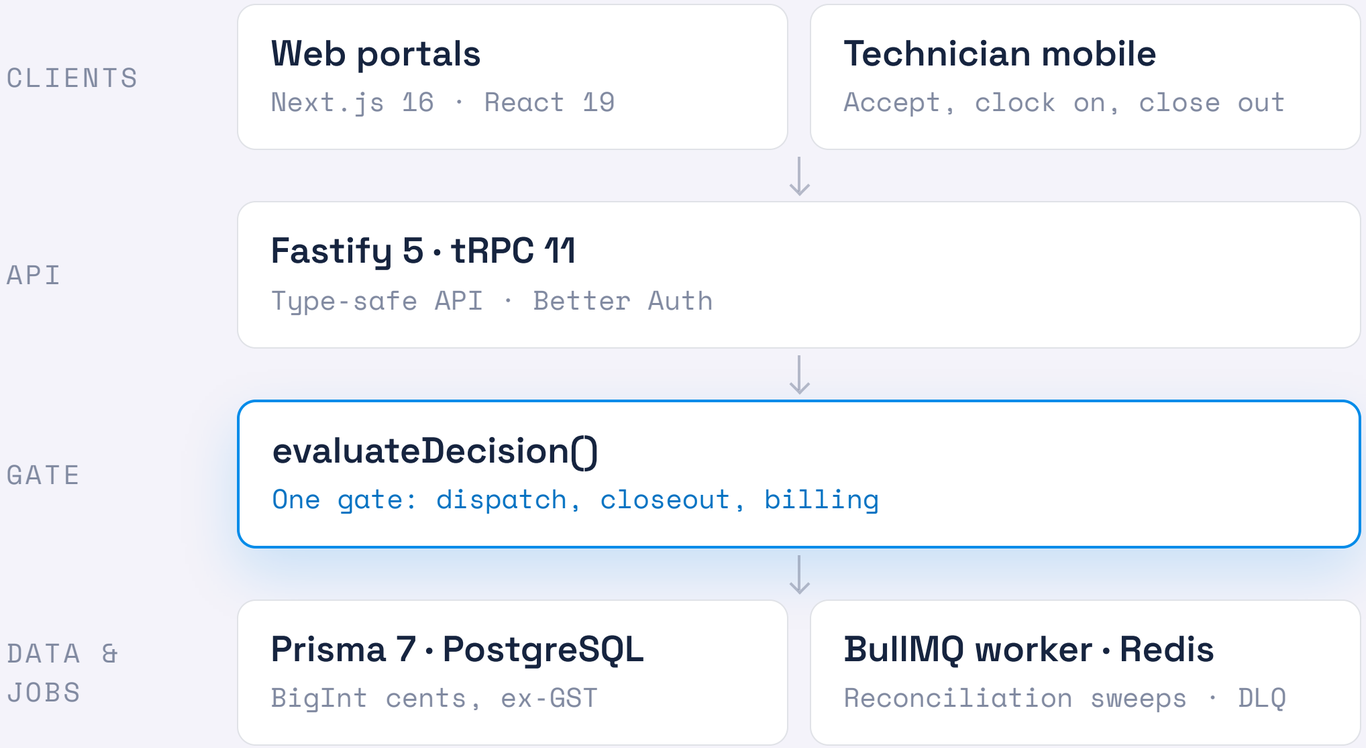
212	1,000+	30K+
PRISMA MODELS	TRPC PROCEDURES	AUTOMATED TESTS

- Five portals — ops, technician, client, subcontractor, public links
- Quote-to-cash, procurement and dual-sided billing behind one gate
- Workforce, assets, compliance and governance — one 212-model schema



01 — ARCHITECTURE & STACK

One enforcement gate behind every workflow



KEY DECISIONS

01 One pure decision engine gates every transition, and every block or override is audited

02 Money as BigInt integer cents, ex-GST: no floating point in any financial path

03 Lint-enforced layering: the UI never imports Prisma, routers never query the DB

04 Event-driven workers with a dead-letter queue and replay tooling

STACK

FRONTEND: Next.js 16 · React 19

BACKEND: Fastify 5 · tRPC 11 · Prisma 7

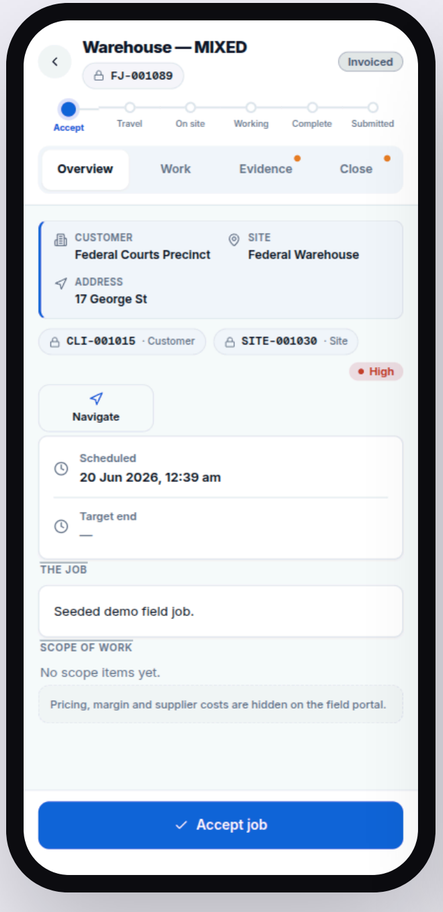
DATA: PostgreSQL · BullMQ · Redis

INFRA: Better Auth · Airwallex · Docker · Railway

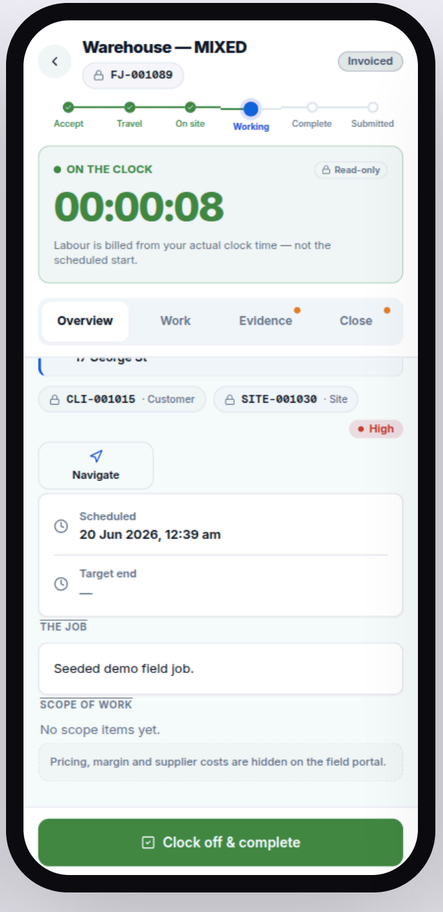
Across admin & field

EVOLUTION DATA & COMMUNICATIONS

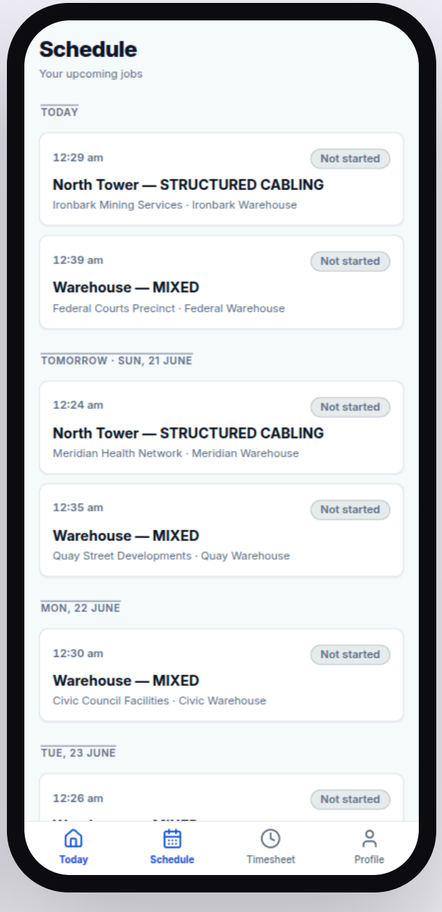
TECHNICIAN APP · MOBILE



Accept job & site brief

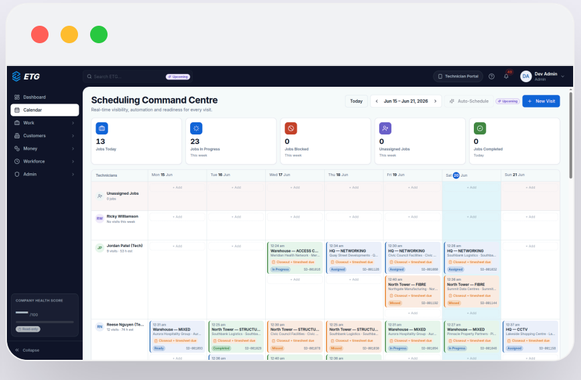


On the clock & evidence

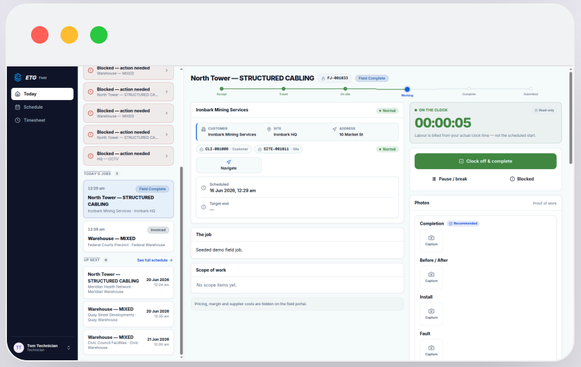


Upcoming schedule

ADMIN & FIELD · WEB



Admin scheduling command centre



Technician web portal: job view

01 — IMPACT

Margin control, enforced

$5k/wk
BILLING LEAKAGE CLOSED

1,000+
TRPC PROCEDURES

30K+
AUTOMATED TESTS

Stops an estimated $5k a week of margin leakage by blocking unbillable or non-compliant work before it ships, backed by 30,000+ tests across 2,200+ test files.

WHAT SHIPPED

- One audited decision gate for dispatch, closeout and billing
- Two-hop bank-feed to supplier-invoice reconciliation, scored
- Every block and override logged with actor and reason
- Production smoke tests gate every release

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